

DAY 1

Calculus:

The Math of Change

Theory: 1 hour **Practical:** 2 hours

Topics: Concept of change • Derivatives • Rules • ML Optimization

What We'll Learn Today

1

0:00 - 1:00

Theory

Concept of change, slopes, derivatives & rules

2

1:10 - 2:10

Practical 1

Guided worked examples on the board

3

2:20 - 3:00

Practical 2

Pair practice + optimization problems

 Don't worry if you feel lost at first — that's completely normal!

P A R

Theory • 1 hour

Let's build intuition step by step

What is Calculus really about?

Calculus is the mathematics of CHANGE.

Everyday examples of change:



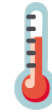
Speed

How fast your position changes over time



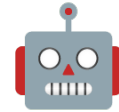
Profit

How sales change as price changes



Temperature

How it changes through the day



AI Learning

How a model's error changes as it learns

You already know slopes!

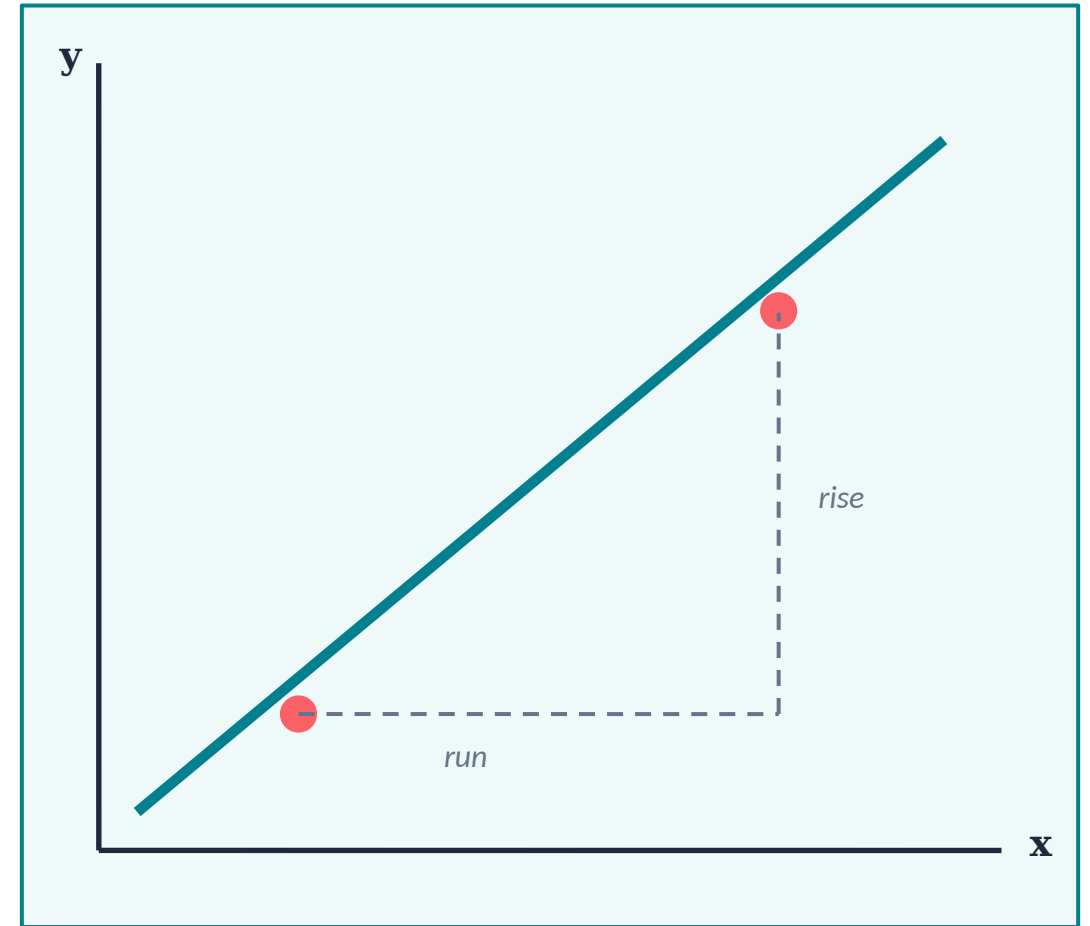
Slope of a straight line:

$$\text{slope} = \text{rise} / \text{run}$$

$$= (\text{change in } y) / (\text{change in } x)$$

It tells us:

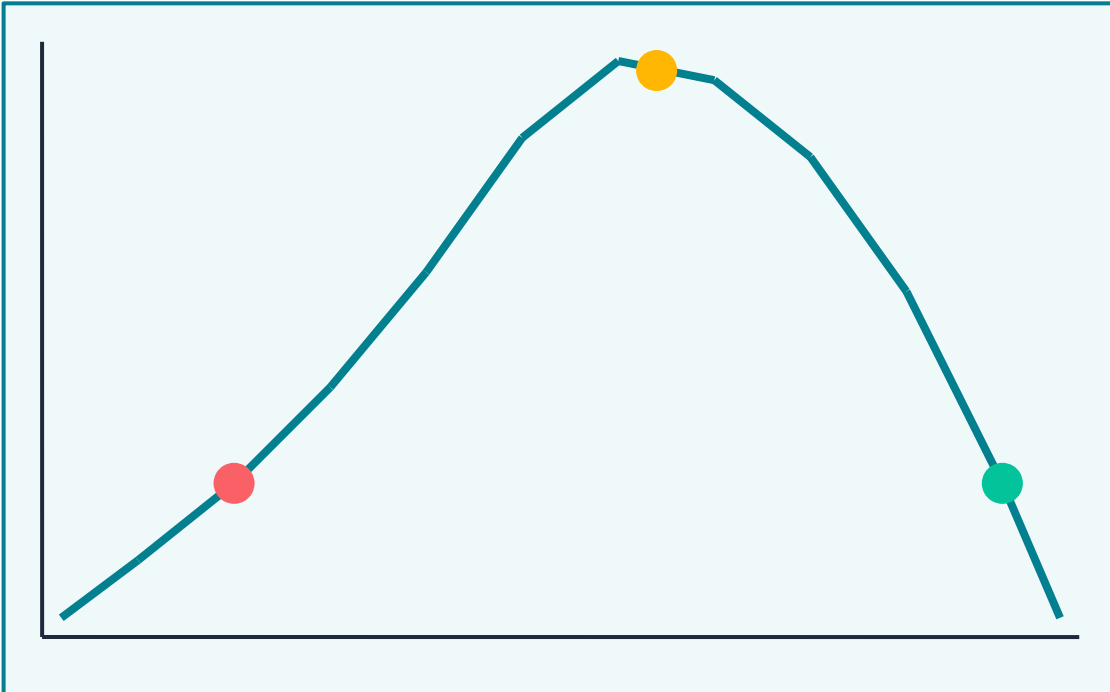
- How steep the line is
- How fast y changes per unit of x
- The direction (up ↗ or down ↘)



But what if it's a curve?

For a straight line, slope is the SAME everywhere.

For a curve, slope **CHANGES** at every point.



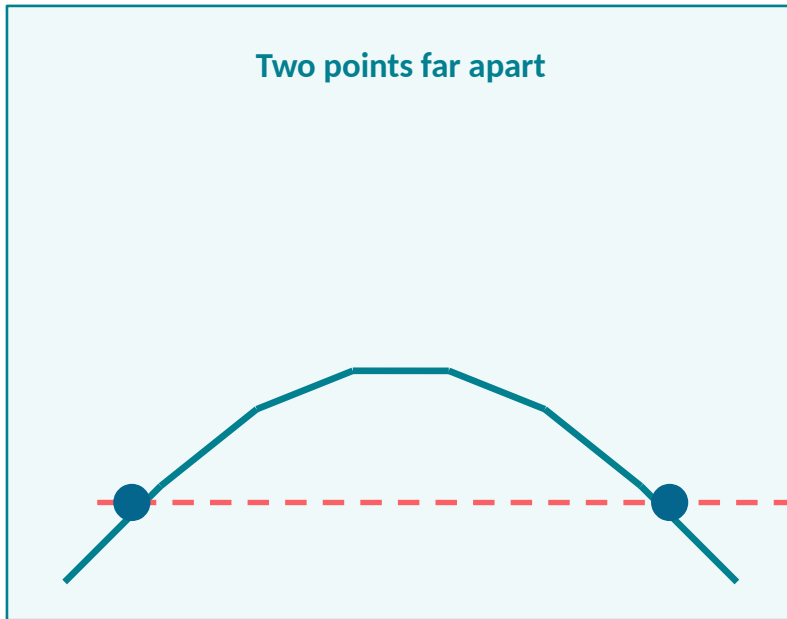
*At every point on a curve,
the slope is different.*

The BIG QUESTION:

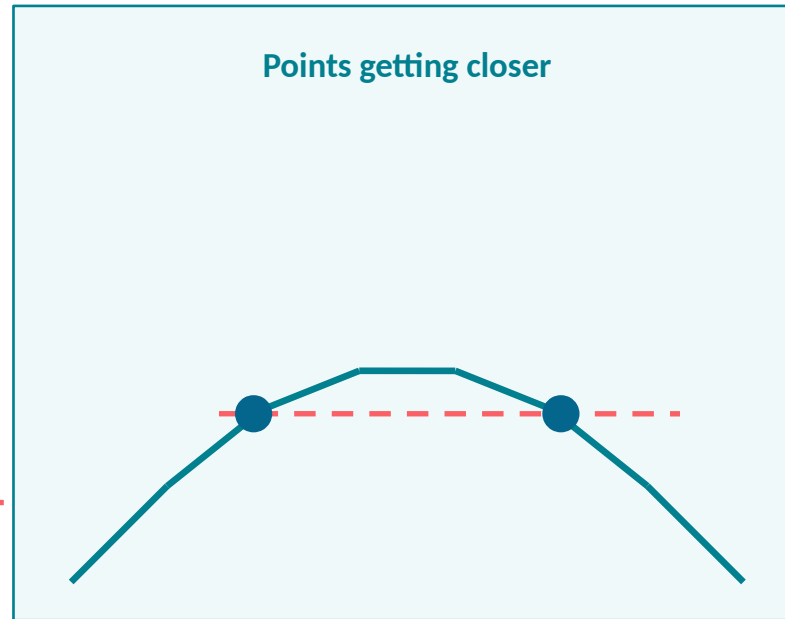
*How do we measure the slope
at a **SINGLE** point on a curve?*

Zoom in closer... and closer...

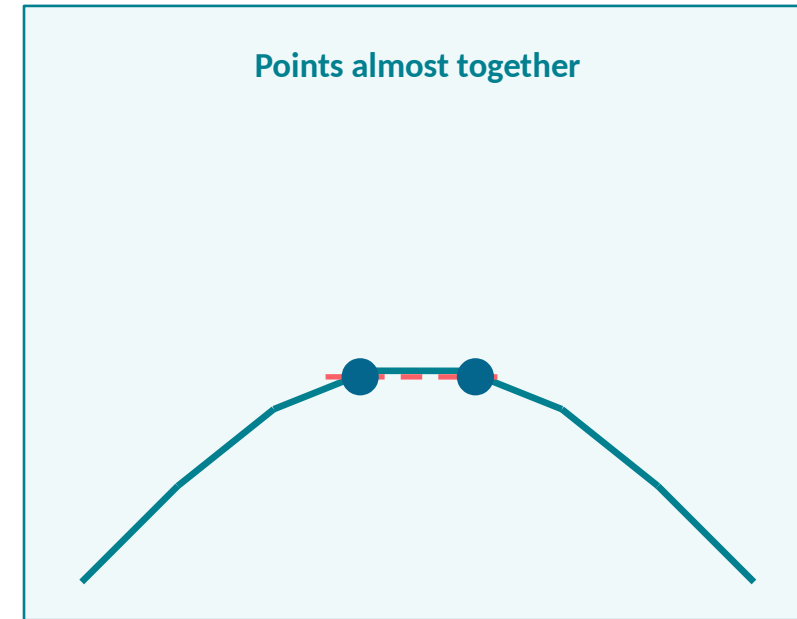
Trick: pick two points on the curve, then move them CLOSER TOGETHER.



rough slope



better slope



TRUE slope at the point!

This limiting slope is called the DERIVATIVE.

The Derivative, Formally

In plain English:

The derivative of $f(x)$ at a point tells us the SLOPE of the curve at that point — in other words, how fast f is changing at that exact moment.

We write it as:

$$f'(x)$$

$$dy/dx$$

$$d/dx [f(x)]$$

Formal limit definition (don't panic — we'll mostly use rules instead):

$$f'(x) = \lim_{\text{as } h \rightarrow 0} [f(x + h) - f(x)] / h$$

Three Ways to Read the Derivative

1

Slope

The slope of the tangent line at point x on the graph of f .



2

Rate of Change

How fast the output changes for a tiny change in input.



3

Velocity

If $f(x)$ is position at time x , then $f'(x)$ is velocity.



5 Rules That Do All the Work

1 Power Rule

$$d/dx [x^n] = n \cdot x^{n-1}$$

2 Sum Rule

$$(f + g)' = f' + g'$$

3 Product Rule

$$(f \cdot g)' = f' \cdot g + f \cdot g'$$

4 Quotient Rule

$$(f/g)' = (f'g - fg') / g^2$$

5 Chain Rule

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

The Most-Used Rule of All

$$\frac{d}{dx} [x^n] = n \cdot x^{n-1}$$

Recipe: bring the exponent down as a multiplier, then subtract 1 from the exponent.

Example

$$f(x) = x^3$$

bring 3 down $\rightarrow 3x^2$, subtract 1 from power

$$f'(x) = 3x^2$$

Example

$$f(x) = x^5$$

bring 5 down $\rightarrow 5x^4$

$$f'(x) = 5x^4$$

Example

$$f(x) = x$$

exponent is 1 $\rightarrow 1 \cdot x^0 = 1$

$$f'(x) = 1$$

★ Bonus: the derivative of a CONSTANT (like 7 or -3) is 0. Constants don't change!

Add Them Up, Differentiate Each

$$(f + g)' = f' + g'$$

In words: differentiate each term separately. The sum rule works for subtraction too (difference rule).

Worked Example

Find $f'(x)$ if $f(x) = x^3 + 5x^2 - 4x + 7$

Step 1: Differentiate $x^3 \rightarrow 3x^2$

Step 2: Differentiate $5x^2 \rightarrow 5 \cdot 2x = 10x$

Step 3: Differentiate $-4x \rightarrow -4$

Step 4: Differentiate $7 \rightarrow 0$ (constant)

$$f'(x) = 3x^2 + 10x - 4$$

Product & Quotient Rules

PRODUCT RULE

$$(f \cdot g)' = f' \cdot g + f \cdot g'$$

Memorize: "first times derivative of second, plus second times derivative of first."

Example: $f(x) = x^2 \cdot \sin(x)$

Let $f = x^2$, $g = \sin(x)$

$f' = 2x$, $g' = \cos(x)$

Apply rule:

$$(x^2 \cdot \sin x)' = 2x \cdot \sin x + x^2 \cdot \cos x$$

QUOTIENT RULE

$$(f / g)' = (f' \cdot g - f \cdot g') / g^2$$

Memory trick: "LO d-HI minus HI d-LO, over LO-squared."

Example: $f(x) = x^2 / (x + 1)$

Let $f = x^2$, $g = x + 1$

$f' = 2x$, $g' = 1$

Apply rule:

$$(2x \cdot (x+1) - x^2 \cdot 1) / (x+1)^2$$

The Function-Inside-a-Function Rule

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

Idea: derivative of the OUTSIDE (keeping inside the same), TIMES derivative of the INSIDE.

Worked Example

Find the derivative of $f(x) = (3x + 5)^4$

Step 1: Identify inside = $3x + 5$. Outside = $()^4$

Step 2: Derivative of outside (power rule on u^4) = $4(3x + 5)^3$

Step 3: Derivative of inside = 3

Step 4: Multiply them together

$$f'(x) = 4(3x + 5)^3 \cdot 3 = 12(3x + 5)^3$$

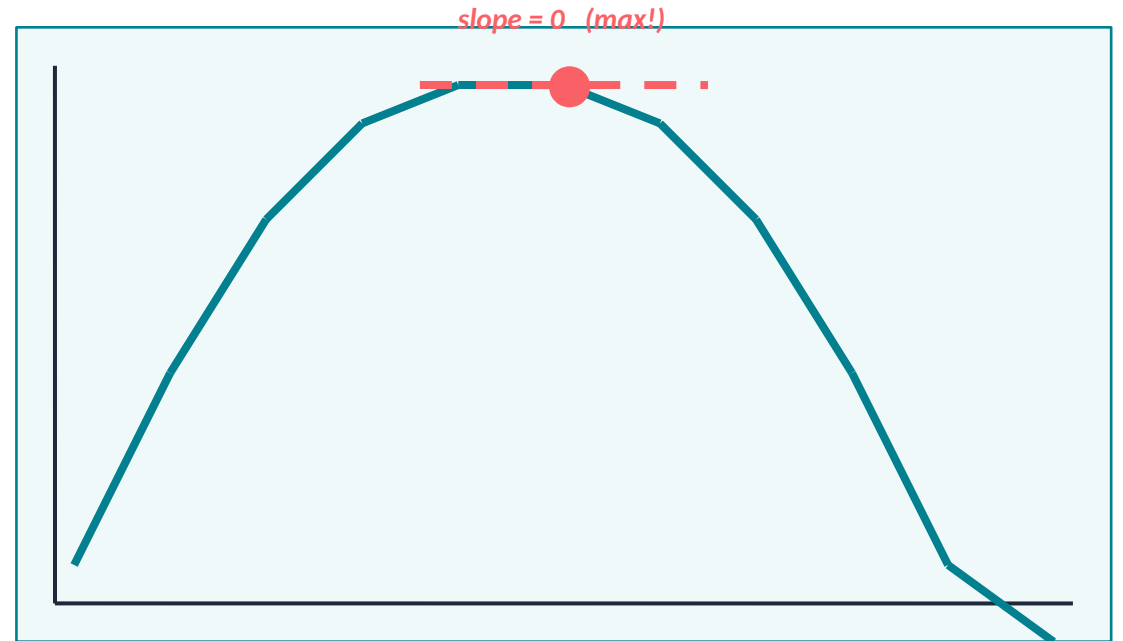
Why Derivatives Matter — Optimization

Optimization = finding the BEST possible value (maximum or minimum)

🔑 **KEY IDEA:** At a maximum or minimum, the slope of the curve is ZERO.
(The tangent line is horizontal — flat.)

The Recipe

1. Write the function you want to maximize/minimize.
2. Find its derivative.
3. Set the derivative = 0.
4. Solve for x . That's your candidate point.
5. Check if it's a max or min.



How AI Uses Derivatives: Gradient Descent



Picture this:

*You're lost in thick fog on a hillside.
You want to reach the lowest valley,
but you can't see 10 meters ahead.*

What do you do?

- Feel the slope beneath your feet.
- Step in the direction it slopes DOWN.
- Repeat, many times.
- Eventually → you reach the valley.



In Machine Learning:

*We want to minimize a LOSS function
(how wrong the model's predictions are).*

Gradient Descent:

- "Slope" = derivative of the loss.
- Step downhill by adjusting weights.
- Repeat thousands of times.
- **The model LEARNS.**



Derivatives are how AI learns!

A Tiny Worked Example

Setup:

Minimize loss $L(w) = w^2$. Start at $w = 4$. Learning rate $\alpha = 0.1$.

Update rule: $w_{\text{new}} = w_{\text{old}} - \alpha \cdot L'(w_{\text{old}})$

Since $L(w) = w^2$, by the power rule $L'(w) = 2w$.

Step	w (current)	$L'(w) = 2w$	Update: $w - 0.1 \cdot L'(w)$	New w
0	4.00	8.00	$4 - 0.1(8) = 4 - 0.8$	3.20
1	3.20	6.40	$3.2 - 0.64$	2.56
2	2.56	5.12	$2.56 - 0.512$	2.05
3	2.05	4.10	$2.05 - 0.41$	1.64
...	...	...	...	...
∞	→ 0	→ 0	—	≈ 0 ✓

✨ We're "walking downhill" toward $w = 0$ — the minimum of $L(w) = w^2$. That's how ML trains!

Theory — What We Learned

1

Derivative = slope

It measures instantaneous rate of change

2

5 rules do it all

Power, sum, product, quotient, chain

3

Slope = 0 at max/min

That's the secret to optimization

4

AI learns via gradients

Gradient descent walks downhill on the loss



Break time — see you in 10 minutes, then we PRACTICE!

P A R

Practical • 2 hours

Let's get our hands dirty with real problems

Worked Example (Together on the Board)

Question:

Find $f'(x)$ where $f(x) = x^7$

Solution:

Step 1: Identify the rule: this is a pure power — use the POWER RULE.

Step 2: Rule says: $d/dx[x^n] = n \cdot x^{n-1}$. Here $n = 7$.

Step 3: Bring the 7 down in front: $7x^{(?)}$

Step 4: Subtract 1 from the exponent: $7 - 1 = 6$.

Step 5: Combine: $f'(x) = 7x^6$

Answer: $f'(x) = 7x^6$

Worked Example (Together on the Board)

Question:

Find $f'(x)$ where $f(x) = 4x^3 + 2x^2 - 5x + 9$

Solution:

Step 1: Use the SUM RULE — differentiate each term separately.

Step 2: $d/dx[4x^3] = 4 \cdot 3x^2 = 12x^2$

Step 3: $d/dx[2x^2] = 2 \cdot 2x = 4x$

Step 4: $d/dx[-5x] = -5$

Step 5: $d/dx[9] = 0$ (constant disappears)

Step 6: Add all pieces together.

Answer: $f'(x) = 12x^2 + 4x - 5$

Worked Example (Together on the Board)

Question:

Find $f'(x)$ where $f(x) = x^2 \cdot (x + 3)$

Solution:

Step 1: Spot a PRODUCT of two functions. Let $u = x^2$, $v = x + 3$.

Step 2: Find $u' = 2x$ and $v' = 1$.

Step 3: Product rule: $(uv)' = u'v + uv'$

Step 4: Substitute: $= 2x \cdot (x + 3) + x^2 \cdot 1$

Step 5: Expand: $2x^2 + 6x + x^2$

Step 6: Combine like terms.

Answer: $f'(x) = 3x^2 + 6x$

Worked Example (Together on the Board)

Question:

Find $f'(x)$ where $f(x) = x / (x + 1)$

Solution:

- Step 1:** Spot a QUOTIENT. Let $u = x$, $v = x + 1$.
- Step 2:** Find $u' = 1$, $v' = 1$.
- Step 3:** Quotient rule: $(u/v)' = (u'v - uv') / v^2$
- Step 4:** Substitute: $= (1(x+1) - x1) / (x+1)^2$
- Step 5:** Simplify numerator: $(x + 1 - x) / (x+1)^2 = 1 / (x+1)^2$

Answer: $f'(x) = 1 / (x + 1)^2$

Worked Example (Together on the Board)

Question:

Find $f'(x)$ where $f(x) = (2x + 1)^5$

Solution:

Step 1: Spot a FUNCTION INSIDE a function. Inside = $2x + 1$. Outside = $()^5$.

Step 2: Derivative of outside (power rule on u^5): $5 \cdot (2x + 1)^4$

Step 3: Derivative of inside $(2x + 1)$: 2

Step 4: Chain rule: multiply them together.

Step 5: $= 5(2x + 1)^4 \cdot 2 = 10(2x + 1)^4$

Answer: $f'(x) = 10(2x + 1)^4$

Worked Example (Together on the Board)

Question:

A farmer has 20 m of fence. Find the max area of a rectangle he can enclose.

Solution:

Step 1: Let width = x . Then length = $(20 - 2x)/2 = 10 - x$. (perimeter = $2x + 2(10 - x) = 20$ ✓)

Step 2: Area $A(x) = x \cdot (10 - x) = 10x - x^2$

Step 3: Find $A'(x) = 10 - 2x$

Step 4: Set $A'(x) = 0 \rightarrow 10 - 2x = 0 \rightarrow x = 5$

Step 5: So width = 5, length = $10 - 5 = 5$. (It's a square!)

Step 6: Max area = $5 \times 5 = 25 \text{ m}^2$

Answer: Max area = 25 m^2 (square, $5 \text{ m} \times 5 \text{ m}$)

Your Turn — Work in Pairs

How this works

- Pair up with the person next to you.
- Work through the practice questions on the next slides.
- Talk out loud — explain your steps to your partner.
- Stuck? Raise your hand. It's OK — we'll help!
- After ~30 min, we'll review every answer together.



Make sure you have paper, a pencil, and the cheat sheet handout ready.

P R A C T I

10 Practice Questions

Solutions follow — try them FIRST, then check!

Questions 1 - 5

Q1

Find $f'(x)$ if $f(x) = x^4$

 Hint: Power rule

Q2

Find $f'(x)$ if $f(x) = 3x^2 + 6x - 8$

 Hint: Sum rule + power rule

Q3

Find $f'(x)$ if $f(x) = 5x^3 - 2x + 1$

 Hint: Sum rule

Q4

Find $f'(x)$ if $f(x) = x \cdot (x^2 + 2)$

 Hint: Expand first, OR use product rule

Q5

Find $f'(x)$ if $f(x) = (x + 1)(x - 3)$

 Hint: Product rule

Questions 6 - 10

Q6

Find $f'(x)$ if $f(x) = (3x - 1) / (x + 2)$



Hint: Quotient rule

Q7

Find $f'(x)$ if $f(x) = (x^2 + 1)^3$



Hint: Chain rule

Q8

Find $f'(x)$ if $f(x) = (5x + 4)^2$



Hint: Chain rule

Q9

Find the x -value where $f(x) = x^2 - 6x + 5$ has a minimum.



Hint: Set $f'(x) = 0$, solve for x

Q10

One ML step: $L(w) = w^2$, $w = 3$, $\alpha = 0.2$. Find new w after one gradient-descent step.



Hint: Use $w_{\text{new}} = w - \alpha \cdot L'(w)$; $L'(w) = 2w$

S O L U

Answer Key

Check your work — and understand every step!

Q1: Power Rule Basic

Problem: $f(x) = x^4$

Step-by-step:

- ▶ Recognize a pure power $\rightarrow$ use POWER RULE: $d/dx[x^n] = n \cdot x^{n-1}$
- ▶ Here $n = 4$.
- ▶ Bring 4 down in front: $4x^{(?)}$
- ▶ Subtract 1 from exponent: $4 - 1 = 3$.
- ▶ Combine: $f'(x) = 4x^3$

✓ Answer: $f'(x) = 4x^3$

Q2: Sum Rule with Power

Problem: $f(x) = 3x^2 + 6x - 8$

Step-by-step:

- ▶ Use SUM RULE — differentiate each term.
- ▶ $d/dx[3x^2] = 3 \cdot 2x = 6x$
- ▶ $d/dx[6x] = 6$
- ▶ $d/dx[-8] = 0$ (constant)
- ▶ Add them: $f'(x) = 6x + 6 + 0 = 6x + 6$

✓ Answer: $f'(x) = 6x + 6$

Q3: Sum Rule — Cubic

Problem: $f(x) = 5x^3 - 2x + 1$

Step-by-step:

- ▶ Apply SUM RULE term by term.
- ▶ $d/dx[5x^3] = 5 \cdot 3x^2 = 15x^2$
- ▶ $d/dx[-2x] = -2$
- ▶ $d/dx[1] = 0$
- ▶ Combine: $f'(x) = 15x^2 - 2$

✓ Answer: $f'(x) = 15x^2 - 2$

Q4: Product — Expand Method

Problem: $f(x) = x \cdot (x^2 + 2)$

Step-by-step:

- ▶ Easiest path: EXPAND first.
- ▶ $f(x) = x^3 + 2x$
- ▶ Now differentiate with sum + power rule.
- ▶ $d/dx[x^3] = 3x^2$, $d/dx[2x] = 2$
- ▶ Result: $f'(x) = 3x^2 + 2$

✓ Answer: $f'(x) = 3x^2 + 2$

Q5: Product Rule

Problem: $f(x) = (x + 1)(x - 3)$

Step-by-step:

- ▶ Let $u = x + 1$, $v = x - 3$.
- ▶ Compute $u' = 1$, $v' = 1$.
- ▶ Product rule: $(uv)' = u'v + uv'$
- ▶ Substitute: $= 1 \cdot (x - 3) + (x + 1) \cdot 1 = (x - 3) + (x + 1)$
- ▶ Simplify: $= 2x - 2$

✓ **Answer:** $f'(x) = 2x - 2$

Q6: Quotient Rule

Problem: $f(x) = (3x - 1) / (x + 2)$

Step-by-step:

- ▶ Let $u = 3x - 1$, $v = x + 2$.
- ▶ Find $u' = 3$, $v' = 1$.
- ▶ Quotient rule: $(u/v)' = (u'v - u v') / v^2$
- ▶ Numerator: $3(x + 2) - (3x - 1) \cdot 1 = 3x + 6 - 3x + 1 = 7$
- ▶ Denominator: $(x + 2)^2$

✓ **Answer:** $f'(x) = 7 / (x + 2)^2$

Q7: Chain Rule — Power of Polynomial

Problem: $f(x) = (x^2 + 1)^3$

Step-by-step:

- ▶ Inside = $x^2 + 1$. Outside = $()^3$.
- ▶ Derivative of outside (power rule on u^3): $3(x^2 + 1)^2$
- ▶ Derivative of inside ($x^2 + 1$): $2x$
- ▶ Chain rule — multiply: $3(x^2 + 1)^2 \cdot 2x$
- ▶ Simplify: $6x(x^2 + 1)^2$

✓ Answer: $f'(x) = 6x(x^2 + 1)^2$

Q8: Chain Rule — Linear Inside

Problem: $f(x) = (5x + 4)^2$

Step-by-step:

- ▶ Inside = $5x + 4$. Outside = $()^2$.
- ▶ Derivative of outside: $2(5x + 4)$
- ▶ Derivative of inside: 5
- ▶ Chain rule: $2(5x + 4) \cdot 5$
- ▶ Simplify: $10(5x + 4) = 50x + 40$

✓ Answer: $f'(x) = 10(5x + 4)$ or $50x + 40$

Q9: Optimization — Find Minimum

Problem: Find the x -value where $f(x) = x^2 - 6x + 5$ has a minimum.

Step-by-step:

- ▶ At a minimum, the slope = 0, so $f'(x) = 0$.
- ▶ Find derivative: $f'(x) = 2x - 6$
- ▶ Set equal to zero: $2x - 6 = 0$
- ▶ Solve: $2x = 6 \rightarrow x = 3$
- ▶ Check: parabola opens upward (coefficient of x^2 is +), so $x = 3$ is a MIN. ✓

✓ **Answer:** Minimum at $x = 3$ (value: $f(3) = 9 - 18 + 5 = -4$)

Q10: Gradient Descent Step

Problem: $L(w) = w^2$, $w = 3$, $\alpha = 0.2$. Find new w after ONE update.

Step-by-step:

- ▶ Derivative of loss: $L'(w) = 2w$ (power rule).
- ▶ At $w = 3$: $L'(3) = 2 \cdot 3 = 6$.
- ▶ Update rule: $w_{\text{new}} = w - \alpha \cdot L'(w)$
- ▶ Substitute: $w_{\text{new}} = 3 - 0.2 \cdot 6 = 3 - 1.2$
- ▶ Simplify: $w_{\text{new}} = 1.8$

✓ **Answer:** New $w = 1.8$ (moved closer to minimum at 0 ✓)



Great work today!

*You now understand what a derivative is,
the 5 rules of differentiation, and how AI uses them.*

Day 2 preview: Integrals — the "opposite" of derivatives.

See you next class! 🙌